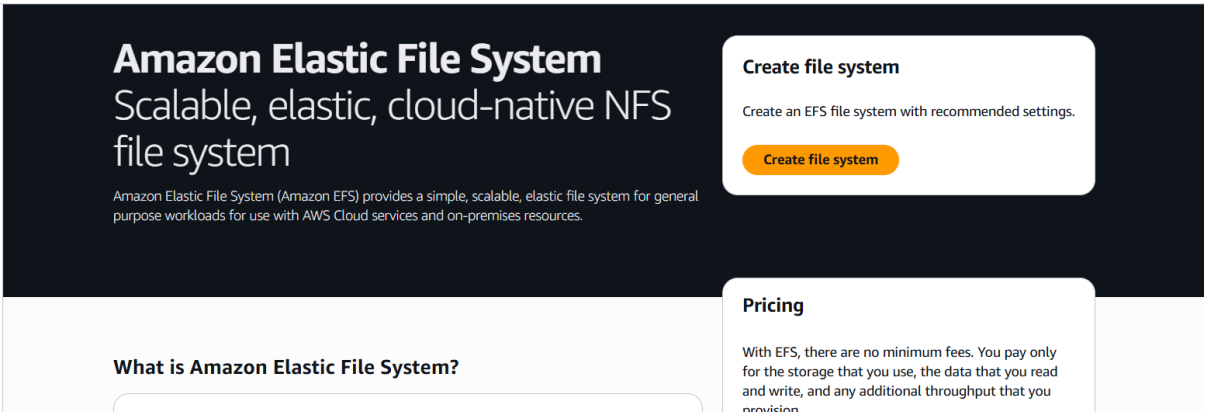
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Task 1: What is Amazon EFS?

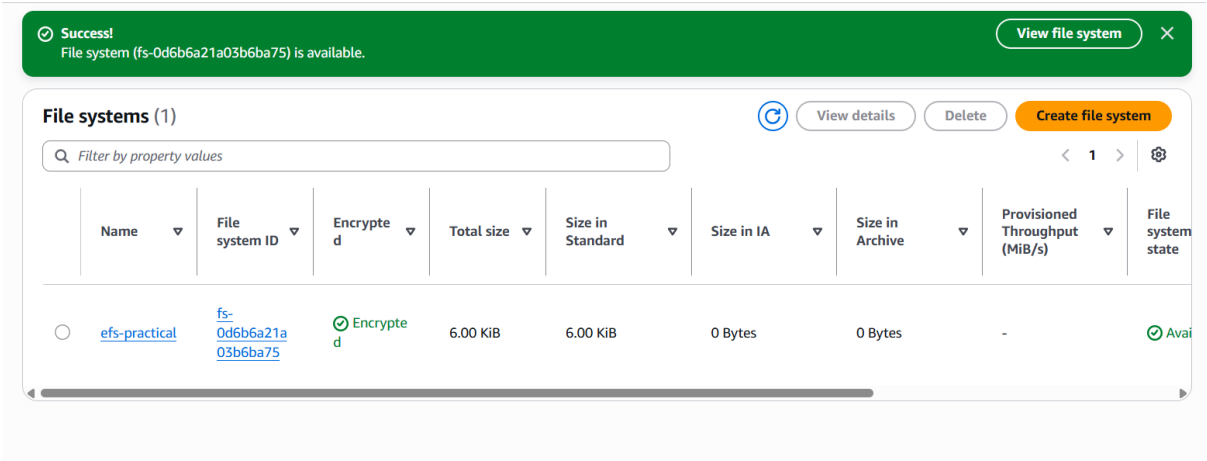
Amazon Elastic File System (EFS) is a scalable, fully managed file storage service provided by AWS.

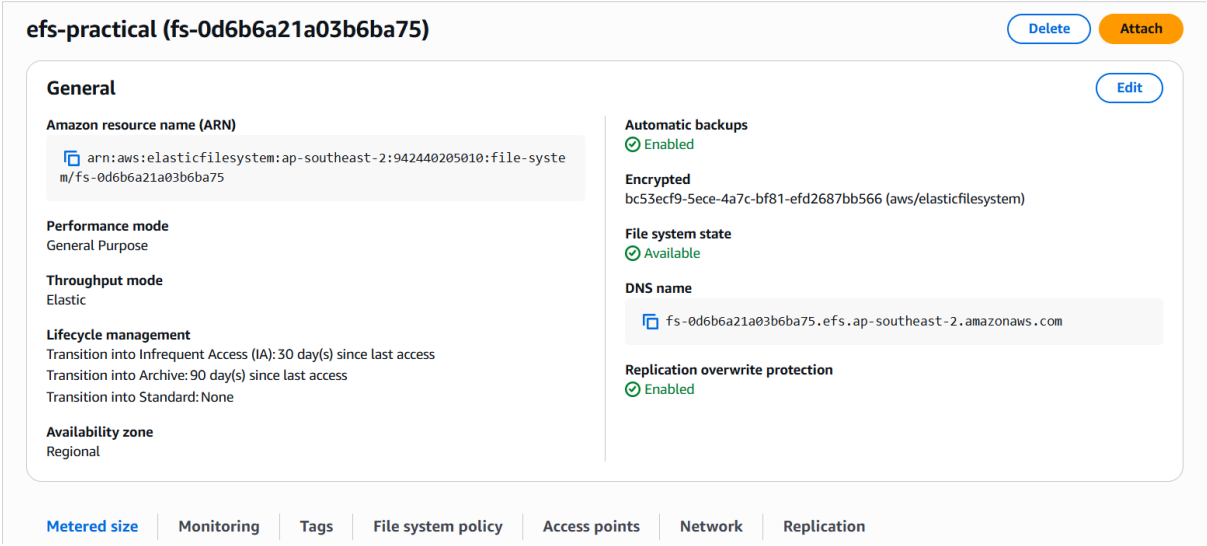
1. EFS allows multiple EC2 instances to access the same files simultaneously.
2. It automatically grows and shrinks as files are added or removed.
3. It provides shared storage through the NFS protocol.
4. EFS supports automatic scaling without requiring manual storage management.
5. It is designed to provide high availability and durability.
6. EFS is commonly used for web servers, shared application files, and content management systems.
7. EFS differs from EBS because EFS provides shared file storage, while EBS provides block storage typically attached to an EC2 instance.

Task 2: Create an Amazon EFS File System

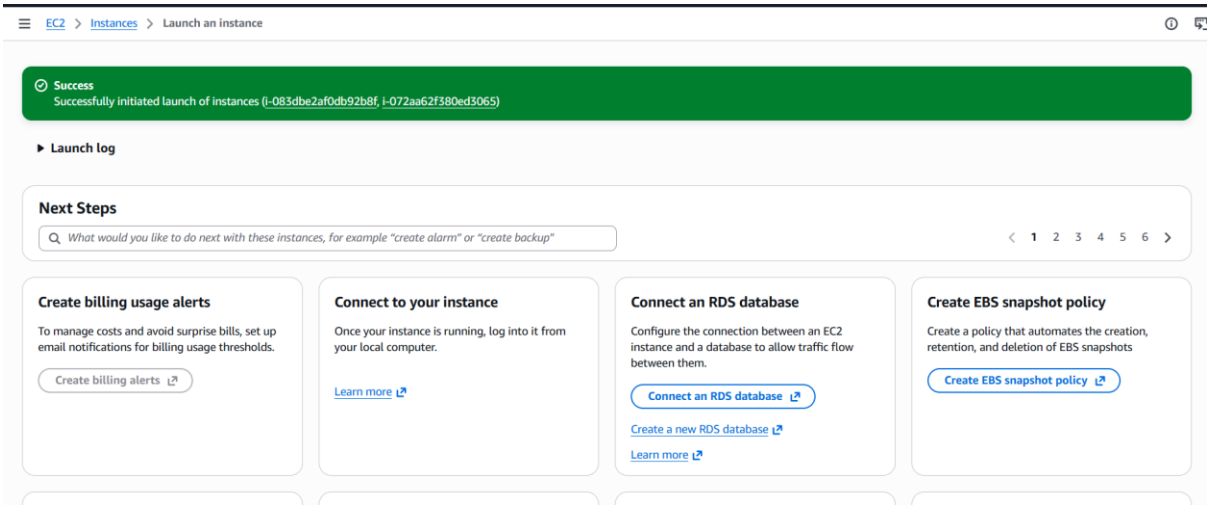


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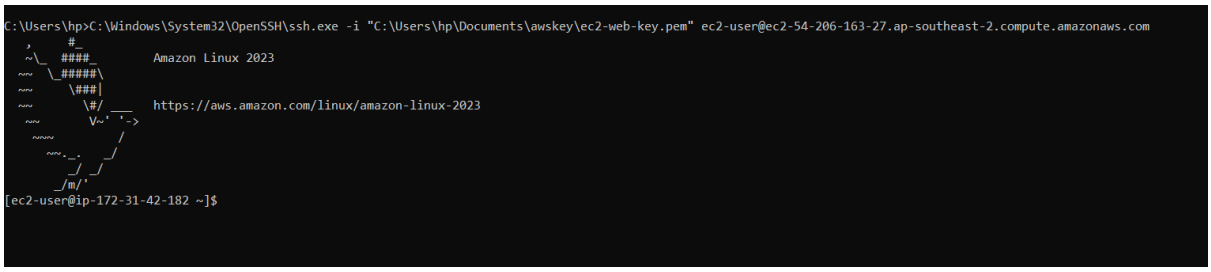


- After that I created 2 instance

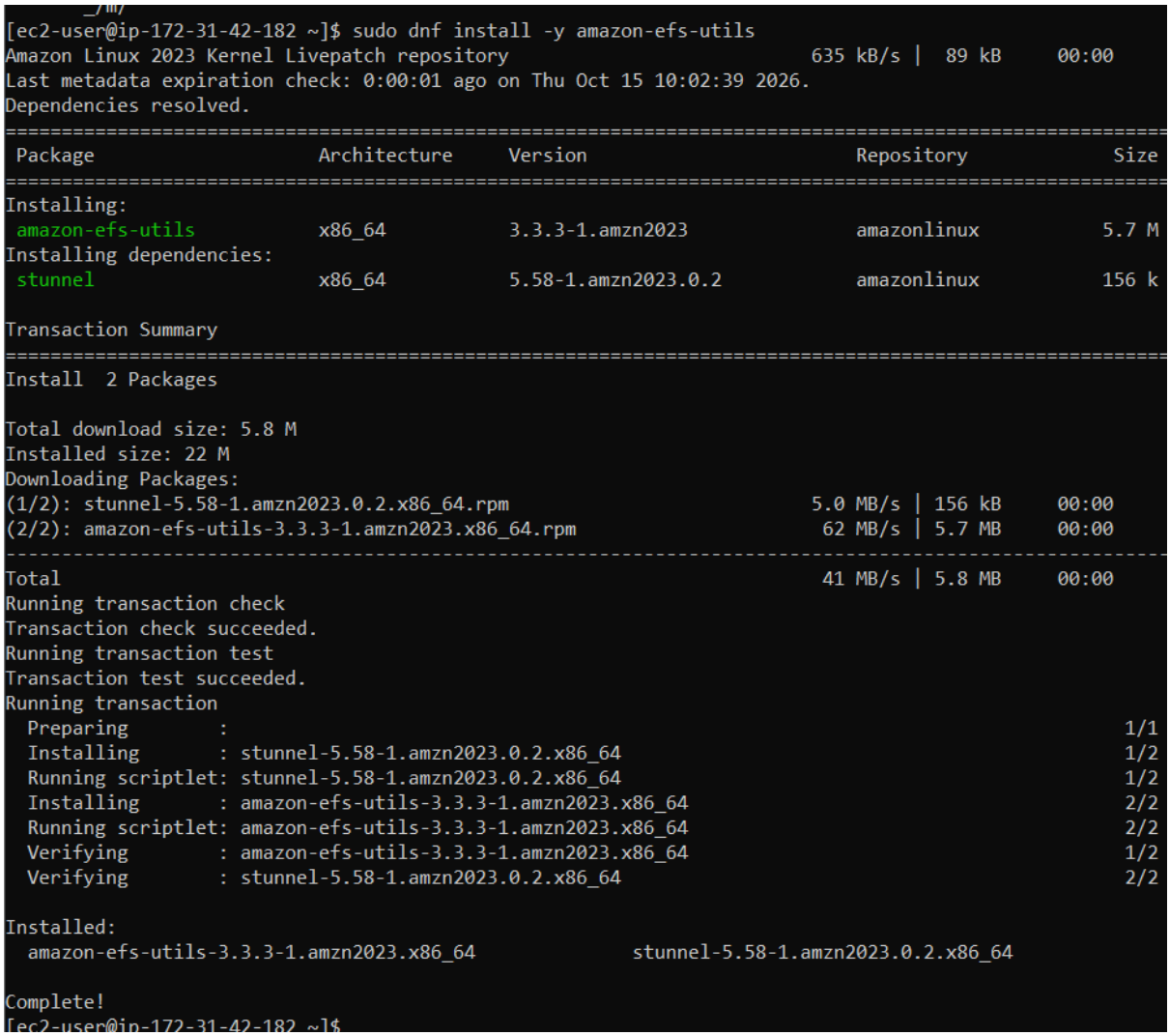


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- Connected through ssh to aws instance



- For server to efs connection I need to install a package called sudo dnf install -y amazon-efs-utils



- First server successfully mounted to efs

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```
C:\Users\hp>\Windows\System32\OpenSSH\ssh.exe -i "C:\Users\hp\Documents\awskey\ec2-web-key.pem" ec2-user@ec2-54-206-163-27.ap-southeast-2.compute.amazonaws.com
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 15 10:35:40 2026 from 111.92.75.43
[ec2-user@ip-172-31-42-182 ~]$ sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport 172.31.9.68:/ /data
^C
[ec2-user@ip-172-31-42-182 ~]$ sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport 172.31.32.164:/ /data
[ec2-user@ip-172-31-42-182 ~]$ lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
[ec2-user@ip-172-31-42-182 ~]$ cd /data/
[ec2-user@ip-172-31-42-182 data]$ echo "welcome to janeeb server" > janeeb.txt
-bash: janeeb.txt: Permission denied
[ec2-user@ip-172-31-42-182 data]$ sudo echo "welcome to janeeb server" > janeeb.txt
-bash: janeeb.txt: Permission denied
[ec2-user@ip-172-31-42-182 data]$ sudo touch janeeb fazil hrishi
[ec2-user@ip-172-31-42-182 data]$
```

- Second server also successfully mounted and connected to efs

```
C:\Users\hp>C:\Windows\System32\OpenSSH\ssh.exe -i "C:\Users\hp\Documents\awskey\ec2-web-key.pem" ec2-user@3.25.130.227
The authenticity of host '3.25.130.227 (3.25.130.227)' can't be established.
ECDSA key fingerprint is SHA256:A8kNyazaB98OolQ3mZjT0JUzdFvYMYYFsfjcEN7zrjg.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.25.130.227' (ECDSA) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-35-23 ~]$ mkdir -p /data
mkdir: cannot create directory '/data': Permission denied
[ec2-user@ip-172-31-35-23 ~]$ sudo mkdir -p /data
[ec2-user@ip-172-31-35-23 ~]$ ^? sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport 172.31.32.164:/ /data sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=104
576,hard,timeo=600,retrans=2,noresvport 172.31.32.164:/ /data
-bash: $'\177': command not found
[ec2-user@ip-172-31-35-23 ~]$  sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport 172.31.32.164:/ /data sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048
6,hard,timeo=600,retrans=2,noresvport 172.31.32.164:/ /data
mount: bad usage
Try 'mount --help' for more information.
[ec2-user@ip-172-31-35-23 ~]$  sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport 172.31.32.164:
mount: 172.31.32.164:: can't find in /etc/fstab.
[ec2-user@ip-172-31-35-23 ~]$  sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport 172.31.32.164:/ /data
[ec2-user@ip-172-31-35-23 ~]$ cd /data
[ec2-user@ip-172-31-35-23 data]$ touch pookie pappu
touch: cannot touch 'pookie': Permission denied
touch: cannot touch 'pappu': Permission denied
[ec2-user@ip-172-31-35-23 data]$ ls
fazil  hrishi  janeeb
[ec2-user@ip-172-31-35-23 data]$
```